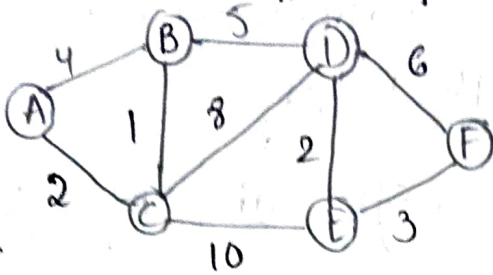


Dijkstra's Algorithm:

Ex-1: shortest paths from A:



priority Queue:

(B, 4)	(C, 2)	(E, 12)	(D, 10)	(F, 14)
(B, 3)		(E, 10)	(D, 8)	(F, 13)
②	①	④	③	⑤

Node	Cost	previous
A	0	-
B	4 3	A, C
C	2	A
D	10 8	C, B
E	12 10	C, D
F	14 13	D, E

at A: $d(A) = 0$

$A \rightarrow B: 0 + 4 < \infty(\checkmark)$ update $d(B) = 4$

$A \rightarrow C: 0 + 2 < \infty(\checkmark)$ update $d(C) = 2$

at C: $d(C) = 2$

$C \rightarrow A: 2 + 2 < 0(\times)$

$C \rightarrow E: 2 + 10 < \infty(\checkmark)$ update

$C \rightarrow D: 2 + 8 < \infty(\checkmark)$ update

$C \rightarrow B: 2 + 1 < 4(\checkmark)$ update

at B: $d(B) = 3$

$B \rightarrow A: 3 + 4 < 0(\times)$

$B \rightarrow C: 3 + 1 < 2(\times)$

$B \rightarrow D: 3 + 5 < 10(\checkmark)$ update

at F: $d(F) = 13$

$F \rightarrow D: 13 + 6 < 8(\times)$

$F \rightarrow E: 13 + 3 < 10(\times)$

at E: $d(E) = 10$

$E \rightarrow C: 10 + 2 < 2(\times)$

$E \rightarrow D: 10 + 2 < 8(\times)$

$E \rightarrow F: 10 + 3 < 14(\checkmark)$ update

now from A to F:

